

University Network Infrastructure Using Multi-Area OSPF

Designed and implemented a scalable campus network architecture using Cisco Packet Tracer. The solution incorporates VLAN segmentation, Inter-VLAN Routing, Multi-Area OSPF, DHCP, DNS, and Web Services to provide reliable communication and centralized network services across multiple departments.

Technologies

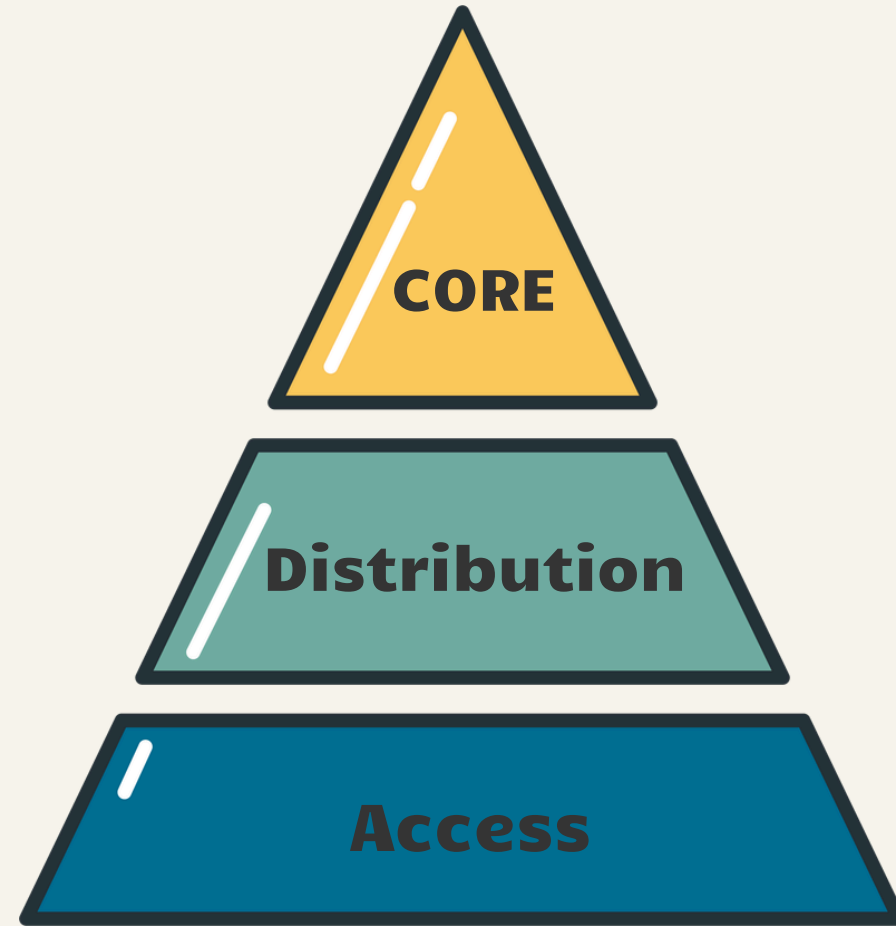
- **VLAN**
- **Trunking**
- **Inter-VLAN Routing**
- **Multi-Area OSPF**
- **DHCP**
- **DNS**
- **HTTP**
- **Layer 3 Switching**

SCOPE

- **Faculty (FIT)**
- **Directorate (DIR)**
- **Library (LIB)**
- **Data Center (DC)**



Network Architecture & Routing Design



Hierarchical Design

Centralized Services	
Service	IP Address
DHCP	192.168.40.10
DNS	192.168.40.11
WEB	192.168.40.12

OSPF Area Design	
Area	Function
Area 0	Backbone
Area 10	Faculty
Area 20	Directorate
Area 30	Library
Area 40	Data Center

ip route

```
Gateway of last resort is not set

10.0.0.0/30 is subnetted, 4 subnets
C    10.0.0.0 is directly connected, FastEthernet0/1
C    10.0.0.4 is directly connected, FastEthernet0/2
C    10.0.0.8 is directly connected, FastEthernet0/3
C    10.0.0.12 is directly connected, FastEthernet0/4
O IA 192.168.10.0/24 [110/2] via 10.0.0.2, 02:11:38, FastEthernet0/1
O IA 192.168.20.0/24 [110/2] via 10.0.0.6, 02:11:00, FastEthernet0/2
O IA 192.168.30.0/24 [110/2] via 10.0.0.10, 02:10:25, FastEthernet0/3
O IA 192.168.40.0/24 [110/2] via 10.0.0.14, 02:10:02, FastEthernet0/4
```

ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
3.3.3.3	1	FULL/DR	00:00:39	10.0.0.6	FastEthernet0/2
4.4.4.4	1	FULL/BDR	00:00:31	10.0.0.10	FastEthernet0/3
5.5.5.5	1	FULL/BDR	00:00:30	10.0.0.14	FastEthernet0/4
2.2.2.2	1	FULL/DR	00:00:31	10.0.0.2	FastEthernet0/1

Testing & Results

DHCP Verification

```
C:\>ipconfig /all

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: 
    Physical Address...                : 000B.BE17.A109
    Link-local IPv6 Address...         : FE80::20B:BEFF:FE17:A109
    IPv6 Address...                    : ::
    IPv4 Address...                    : 192.168.10.100
    Subnet Mask...                     : 255.255.255.0
    Default Gateway...                 : ::
                                        192.168.10.1
    DHCP Servers...                   : 192.168.40.10
    DHCPv6 IAID...                     : 
    DHCPv6 Client DUID...              : 00-01-00-01-40-2A-D3-BD-00-0B-BE-17-A1-09
    DNS Servers...                     : ::
                                        192.168.40.11
```

DNS Resolution Verification

```
C:\>ping www.telkomuniversity.local

Pinging 192.168.40.12 with 32 bytes of data:

Reply from 192.168.40.12: bytes=32 time=20ms TTL=125
Reply from 192.168.40.12: bytes=32 time<1ms TTL=125
Reply from 192.168.40.12: bytes=32 time<1ms TTL=125
Reply from 192.168.40.12: bytes=32 time=1ms TTL=125

Ping statistics for 192.168.40.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 20ms, Average = 5ms
```

HTTP Verification

